



New indications of potential rat intestinal α -glucosidase inhibition by *Syzygium zeylanicum* (L.) and its hypoglycemic effect in mice

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Abstract

Natural α -glucosidase inhibitors (aGIs) have been considered effective agents for type 2 diabetes management. In this study, *Syzygium zeylanicum* (L.) (SZL), a medicinal plant that grows wild in the Central Highlands of Vietnam, was reported for the first time to have potential anti-diabetic properties. The methanol extract of SZL possessed potent rat intestinal α -glucosidase inhibitory activity with low IC₅₀ and great maximum inhibition values of 109 μ g/mL and 100%, respectively. This was higher activity than acarbose, an anti-diabetic drug with IC₅₀ and maximum inhibition values of 540 μ g/mL and 80%, respectively. In specific inhibition tests, the herbal extract demonstrated strong inhibition against α -glucosidases from *Saccharomyces cerevisiae* and *Bacillus stearothermophilus* with small IC₅₀ values of 0.18 and 0.31 μ g/mL, respectively. SZL extract also demonstrated potent inhibition

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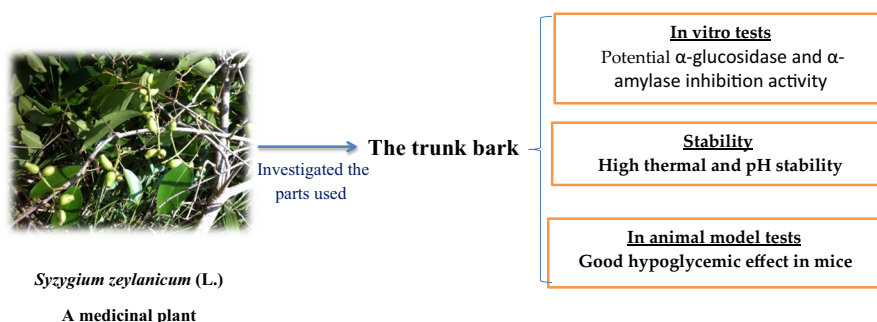
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against porcine pancreatic α -amylase with maximum inhibition and IC₅₀ values of 99.2% and 3.7 μ g/mL, respectively. SZL displayed high thermal and pH stability, with respective activities of 87% and 100–140% when treated at 100 °C or over a wide pH range of 2–13. The herbal extract was further investigated for its efficacy using ICR mice as the animal model. At a dose of 100 mg/kg BW, SZL significantly reduced blood glucose in mice and no signs of diarrhea were observed. The results from this study suggest that SZL, a potent and natural source of aGIs, may be a good candidate for the treatment of type 2 diabetes.

Graphic abstract

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Keywords α -Glucosidase inhibitors · Diabetes · *Syzygium zeylanicum* (L.) · Hypoglycemic effect

Introduction

Type 2 diabetes (T2D) is recognized as a global health issue. By 2030, it is estimated 330 million people worldwide will be affected by T2D [1]. Alpha-glucosidase inhibitors (aGIs) have been extensively studied for T2D management [2, 3]. These molecules are obtained from various sources, including medicinal plants [4–11], chemical synthesis [3, 4, 12], microbial fermented products [13–20], marine organisms [21–24], and fungal products [25–30]. Of these, medicinal plants, microbial fermentation, and chemical synthesis are the major sources of aGIs. Chemically synthesized aGIs have been reported to cause side effects [5]. Microbes can produce numerous bioactive products [19, 31–37], including aGIs [13]. Like microbial sources, medicinal plants are also abundant in bioactive metabolites. Of the major aGIs sources, herbal aGIs have been of great interest due to their nature and since they are safe to use [7, 9].

Medicinal plants have been used worldwide as folk medicine, especially in Asian countries like Vietnam [6]. Vietnam is a tropical country, currently ranked the

sixteenth most biodiverse, with more than 10,000 reported plant species and around 4000 herbal species that have been used as medicinal folk medicine [38]. The study of bioactive herbs has significantly increased in recent years [7–9, 18–20]. However, there have been few new records of medicinal plants from this biodiverse land with potential anti-diabetic properties [6].

Syzygium zeylanicum (L.) (SZL), a medicinal plant, grows wild in the Central Highlands of Vietnam. This herb also grows in other countries, including China, India, Sri Lanka, Bangladesh, Myanmar, Thailand, Malaysia, Indonesia, and Madagascar [39]. Several *Syzygium* species, especially *Syzygium polyanthum* (Wight) Walp, have been reported as possessing anti-diabetic, antioxidant, and toxic activities [40]. Cytotoxic and antimicrobial activities as well as phytochemicals were also reported for *Syzygium polyanthum* [41]. In contrast, SZL has rarely been investigated for its biological activities.

This study first investigated the anti-diabetic activity of SZL. The trunk bark of SZL was collected in the Central Highlands of Vietnam and evaluated for its inhibition of α -glucosidase in rat intestines and its hypoglycemic effect in mice. The specific enzyme inhibition and the thermal and pH stability were also investigated. This study contributed to the catalog of novel anti- α -glucosidase and hypoglycemic medicinal plants.

Experimental

Materials

The medicinal plant sample (*Syzygium zeylanicum* L.) was collected in the Dak Lak province of the Central Highlands of Vietnam; ELC extract was from our previous study [5]. Porcine pancreatic α -amylase (type VI-B) was purchased from Sigma-Aldrich, St. Louis, MO, USA. Acarbose, *Saccharomyces cerevisiae* α -glucosidase, and *Bacillus stearothermophilus* α -glucosidase were purchased from Sigma Chemical Co., St. Louis, MO, USA. Rat α -glucosidase (intestinal acetone powders from rat) was purchased from Sigma-Aldrich, Singapore. When possible, reagents, solvents, and other common chemicals were obtained at the highest grade available.

Extraction method

All samples were dried at ambient temperature and ground to powder (10 g). The powder was extracted for 24 h at room temperature with 100 mL solvent (water, butanol, ethyl acetate, methanol, and ethanol) in a glass conical flask using a shaker and then filtered (filter paper No. 1, Whatman International Ltd., Maidstone, England). The residues were extracted twice with 200 mL solvent, as described above. The combined extracts were concentrated using a rotary evaporator (IKA, Germany) at 60 °C under vacuum conditions. The extracts were stored at –30 °C for later use [5]. Highest activity occurred when the sample was extracted by methanol; therefore, this extract was chosen for further animal tests.

Enzymatic inhibitory assay

The α -glucosidase inhibition assay was performed as per the methods previously described [18]. In brief, a mixture of 50 μ L of sample solution, 50 μ L of α -glucosidase, and 100 μ L potassium phosphate buffer was pre-incubated for 20 min at 37 °C. The reaction was started by adding 50 μ L of the substrate *p*-nitrophenyl glucopyranoside (*p*NPG) and maintained at 37 °C for 40 min. The reaction was stopped by the addition of 100 μ L of 1 M Na₂CO₃, and the absorption of the final mixture was measured at 410 nm. α -Glucosidase inhibition was calculated according to the formula:

$$\text{aGI (\%)} = (A - B)/A \times 100$$

where aGI (%) is α -glucosidase inhibitory activity (%), *A* is the absorbance at 410 nm of the control group (reaction mixture without inhibitor/sample), and *B* is the absorbance of the experimental group (the reaction mixture with inhibitor/sample). The IC₅₀ value, defined as the inhibitor concentration that inhibits 50% of enzymatic activity under test conditions, was calculated using the equation obtained from 3 to 5 linear plots of the samples (inhibitors). 0.1 mol/L potassium phosphate buffer (pH 7) was used to prepare the enzyme solutions, including *Bacillus stearothermophilus* and *Saccharomyces cerevisiae* α -glucosidases at concentrations of 0.10 and 0.25 U/mL, respectively. Rat intestinal α -glucosidase solution was prepared according to the previous report [13]. α -Amylase inhibition was detected using the methods described by Nguyen et al. [7].

Experimental animal protocol

The test for hyperglycemic activity was performed according to the methods described in detail in the previous work [3, 13]. Male ICR mice aged 48 weeks were obtained from The National Laboratory Animal Center (Taipei City, Taiwan) and randomly divided into six groups (eight mice per group). All groups had equivalent mean plasma glucose levels and body weights. Three experimental groups were given oral SZL extract doses of 50, 100, or 200 mg/kg BW, two experimental groups received oral acarbose (12.5 or 25 mg/kg), and the final group (control) received only distilled water. All tests using ICR mice as animal models were carefully conducted in accordance with the guidelines and approval of the Institutional Animal Care and Use Committee of the National Research Institute of Chinese Medicine, Ministry of Health and Welfare (IACUC No.104-706-1, 29/12/2014).

ICR mice in all groups fasted for 16 h and were then given distilled water (control group), acarbose, or SZL extract (experimental groups). After 25 min of sample loading, a 3 g/kg BW sucrose solution was orally administered. Blood was regularly sampled and measured 0.5, 1, 1.5, and 2 h after administration of sucrose solution.

Statistical analysis

Statistical Analysis Software (SAS 9.4) provided by the SAS Institute Taiwan Ltd. (Taipei City, Taiwan) was used to analyze the significant differences of the

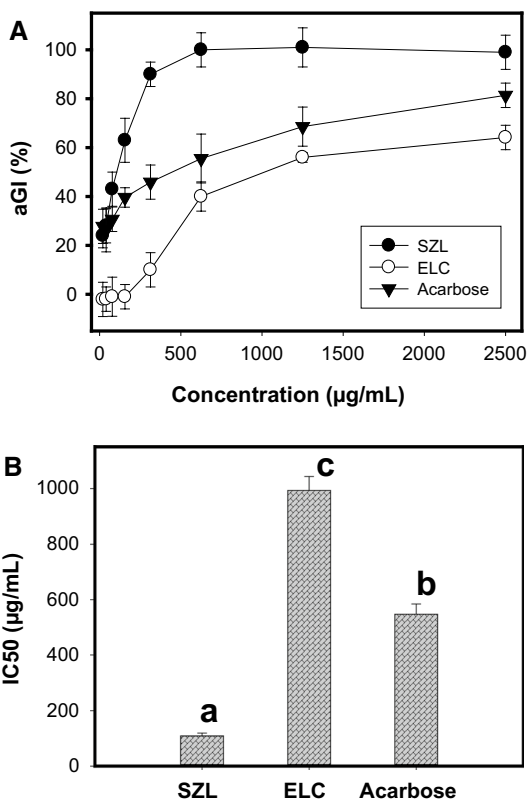
calculated IC_{50} values, the α -glucosidase inhibition (%), and the plasma glucose levels.

Results and discussion

Syzygium zeylanicum L. extracts as potential sources of rat intestinal aGIs

Syzygium zeylanicum L., an herb widely distributed in the Central Highlands of Vietnam and other Asian countries [39], is used in folk medicine to manage several diseases, including diabetes [42]. However, no scientific proof exists for this effect. In the current study, we used α -glucosidase inhibitory activity as indicator of the in vitro anti-diabetic effect of SZL. The trunk bark of SZL was extracted by methanol and then tested for α -glucosidase inhibitory activity. As the results illustrated in Fig. 1, the SZL extract demonstrated potent rat intestinal α -glucosidase inhibitory activity with maximum inhibition (Fig. 1a) and IC_{50} values (Fig. 1b) of 100% and 109 μ g/mL, respectively. The activity of SZL is significantly higher

Fig. 1 The α -glucosidase inhibitory activity of the methanol extracts from the medicinal plants *S. zeylanicum* L. (SZL) and *E. laxiflorus* Champ (ELC) and the positive control acarbose. Rat α -glucosidase was used as the enzymatic source, and inhibition was expressed as % (a) and IC_{50} value (b). Means with the same letter are not significantly different in b based on Duncan's multiple range test ($\alpha=0.01$). $LSD=158.13$; $CV\%=7.652537$



than that of acarbose, a commercial α -glucosidase inhibitor, which had maximum inhibition and IC₅₀ values of 80% and 540 μ g/mL, respectively.

In our previous study, the medicinal plant *Euonymus laxiflorus* Champ (ELC) was the most effective α -glucosidase inhibitor source among the various herbs collected in the Central Highlands of Vietnam [5, 11]. The bark of this herb was also tested in this study for comparison and demonstrated moderate maximum inhibition and IC₅₀ values of 60% and 1000 μ g/mL, respectively. Recently, rat intestinal α -glucosidase inhibitory activity was reported for the heartwood extract of *Dalbergia tonkinensis*, as well as moderate maximum inhibition and IC₅₀ values of 61% and 1720 μ g/mL [6]. Based on recent literature, not only is this the first report of rat intestinal α -glucosidase inhibitory activity in SZL, the extract also displayed the most effective inhibition against rat intestinal α -glucosidase compared to other medicinal plants recently collected from the same area [5, 6, 11].

The trunk bark of SZL was extracted by various solvents, including water, butanol, ethyl acetate, methanol, and ethanol, and then tested for activity. As shown in Fig. 2, the samples extracted by water, methanol, and ethanol displayed stronger activity (ranked at D level) than samples extracted by butanol and ethyl acetate (ranked at A and B level, respectively) or acarbose (ranked at C level). Of these, the methanol extract had the lowest IC₅₀ value at 106 μ g/mL and, as such, demonstrated the most effective inhibition against α -glucosidase. Therefore, this extract was used for subsequent experiments.

Specific inhibitory activity of SZL extract against some enzymes

It is well known that amylase degrades polysaccharides into dextrin oligomers, and α -glucosidase further degrades these oligomers to α -D-glucose before the monomer

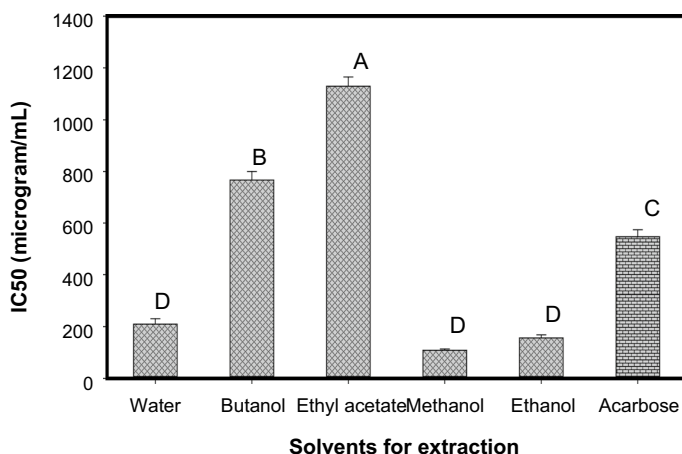


Fig. 2 The α -glucosidase inhibitory activity of different extracts from SZL. The trunk bark of SZL was extracted with water, butanol, ethyl acetate, methanol, and ethanol. The inhibition was tested against rat α -glucosidase and expressed as IC₅₀ values. Means with the same letter are not significantly different in the figure based on Duncan's multiple range test ($\alpha=0.01$). CV%=9.205507; LSD = 115.59

sugar enters blood circulation. Therefore, both α -glucosidase and α -amylase inhibitors are useful for their hypoglycemic and anti-diabetic effects [5, 7]. To determine its specific enzymatic inhibition and confirm SZL may be a potent anti-diabetic drug, five kinds of enzymes were tested, including three α -glucosidases and two α -amylases (Table 1).

The results given in Table 1 indicate that SZL strongly inhibits *S. cerevisiae* α -glucosidase and *B. stearothersophilus* α -glucosidase but only moderately inhibits rat α -glucosidase, with IC₅₀ values of 0.18, 0.31 and 109 μ g/mL, respectively. Acarbose also strongly inhibits *B. stearothersophilus* α -glucosidase and moderately inhibits rat α -glucosidase but shows quite weak inhibition of *S. cerevisiae* α -glucosidase with IC₅₀ values of 0.022, 547 and 1495 μ g/mL, respectively. While SZL displayed weaker inhibition of *B. stearothersophilus* α -glucosidase, it had much stronger inhibition of rat α -glucosidase and *S. cerevisiae* α -glucosidase than acarbose. SZL also demonstrated good inhibition against porcine pancreatic α -amylase and *B. subtilis* with low IC₅₀ and high maximum inhibition values of 3.7–12.7 μ g/mL 93–99.2%, respectively. Based on these results, SZL may be a good candidate for both α -glucosidase and α -amylase inhibition and may have useful hypoglycemic and anti-diabetic effects.

pH and thermal stability of SZL extract

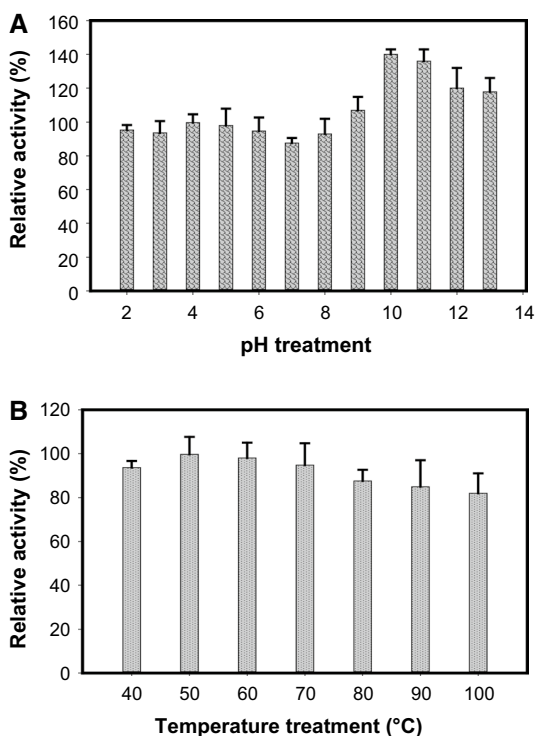
It has been suggested that a potent α -glucosidase inhibitor should display pH stability, especially in acidic conditions since it is often acidic in the gastrointestinal tract [5, 6, 13, 14]. Therefore, the pH stability of SZL was tested. As shown in Fig. 3a, SZL inhibitors demonstrated very high pH stability since $\geq 90\%$ of relative activity was retained. Relative activity even reached 105–140% when SZL was treated between pH 9 and 13. Compared to other natural product extracts, the pH stability of SZL was comparable to that of squid pens and nutrient broth fermented by *Paenibacillus* sp. (relative activity $\geq 87\%$) [13, 14], slightly higher than that of fermented shrimp heads (relative activity of 80–95%) or *Dalbergia tonkinensis* heartwood extract (relative activity of 80–98%), but significantly higher than that of shrimp heads fermented by *Bacillus mycoides* and *Rhizobium* sp. (relative activity $> 60\%$)

Table 1 Specific inhibitory activity of SZL and acarbose against enzymes

Enzyme	Inhibition of SZL		Inhibition of acarbose	
	IC ₅₀ (μ g/mL)	Maximum inhibition (%)	IC ₅₀ (μ g/mL)	Maximum inhibition (%)
<i>S. cerevisiae</i> α -glucosidase	0.18 $\pm$ 0.01	93 $\pm$ 2.2	1495 $\pm$ 10	64 $\pm$ 2.3
Rat α -glucosidase	109 $\pm$ 6.1	100 $\pm$ 4.1	547 $\pm$ 2.5	8 $\pm$ 3.4
<i>B. stearothersophilus</i> α -glucosidase	0.31 $\pm$ 0.07	98 $\pm$ 1.6	0.022 $\pm$ 0.001	100 $\pm$ 2.2
Porcine pancreatic α -amylase	3.7 $\pm$ 0.21	99.2 $\pm$ 1.8	5.2 $\pm$ 0.09	91 $\pm$ 2.5
<i>B. subtilis</i> α -amylase	12.6 $\pm$ 1.11	93 $\pm$ 2.1	ND	ND

ND not determined

Fig. 3 The pH (a) and thermal (b) stability of SZL was determined by treating SZL in the pH range of 2–13 and temperature range of 40–100 °C for 30 min. aGI activity was tested and expressed as relative activity (%) under the same conditions using the enzymatic inhibition assay mentioned in “Methods” section. Yeast α -glucosidase was used for enzymatic inhibition assay



[15] or ELC extract (relative activity of 48–55% at pH 4–8) [5]. In addition, SZL also possessed high thermal stability, with 80–100% relative activity even when treated in the range of 40–100 °C for 30 min (Fig. 3b). These results indicate that SZL is a valuable source of α -glucosidase inhibitors due to its potential activity and high stability.

Effect of SZL extract on plasma glucose in mouse model

To evaluate SZL's effect on plasma glucose, the sucrose tolerance assay [3, 13] was used. ICR mice were used as animal models to determine the hypoglycemic effect. SZL was tested at doses of 50, 100, or 200 mg/kg BW (body weight). Acarbose was also tested for comparison as a control. The results of the animal tests are illustrated in Fig. 4. At a dose of 50 mg/kg, SZL reduced plasma glucose 0.5–1 h after loading. At doses of 100–200 mg/kg, SZL had a more significant hypoglycemic effect. The hypoglycemic effect in mice given 100 mg/kg SZL is comparable to that in mice dosed with 12.5 mg/kg acarbose, but slightly lower than those dosed with 25 mg/kg acarbose.

Several natural products have been reported to reduce plasma glucose in the same mouse model (ICR mice). Fermented shrimp shells significantly reduced blood glucose in ICR mice at a dose of 100 mg/kg [3], while fermented nutrient

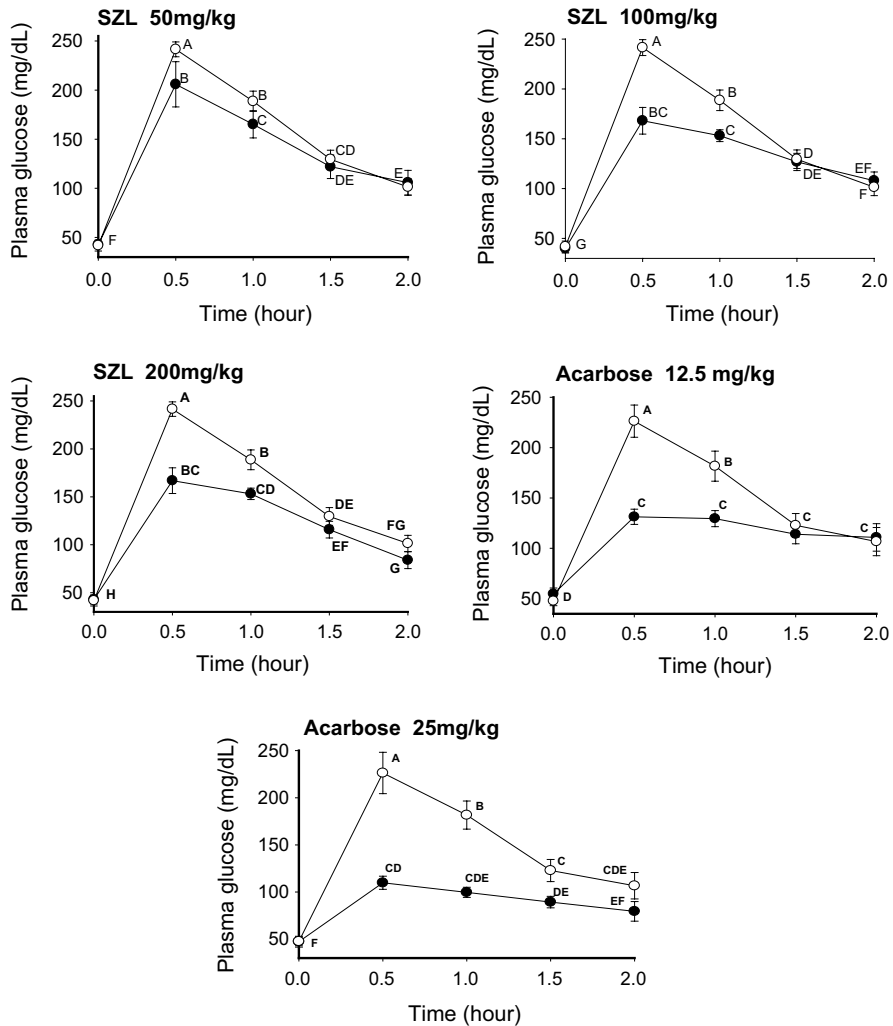


Fig. 4 Effects of SZL and acarbose on plasma glucose levels following oral sucrose loading in ICR mice. SZL (50, 100, and 200 mg/kg BW) and acarbose (12.5 and 25 mg/kg BW) were orally administered to mice (●, $n=8$). In the control group (○, $n=8$), distilled water was administered. After loading the sample (or acarbose or distilled water) for 25 min, sucrose solution was orally administered at a concentration of 3 g/kg BW. Blood was sampled and measured 0.5, 1, 1.5, and 2 h after loading. Different letters in the same sub-figure indicate significant differences between plasma glucose values, according to Duncan's multiple range test ($\alpha=0.05$)

broth had a good effect on glucose levels at high doses of 200–400 mg/kg [13]. Recently, a component rich in condensed tannin was reported to show high glycaemic effect in ICR mice at dose of 100 mg/kg [8]. In this study, 100 mg/kg SZL also displayed high glycaemic effect, suggesting this herbal extract may be a valuable source of hypoglycaemic compounds.

In the previous reports [3, 13], we reported that fermented shrimp shells and fermented nutrient broth reduced plasma glucose due to their high α -glucosidase inhibition, while condensed tannin [8] demonstrated high glycemic properties since it could inhibit both α -glucosidase and α -amylase. Like condensed tannin, SZL showed quite high α -glucosidase and α -amylase inhibitory activity; therefore, it could have a great effect on reducing blood glucose levels in mice.

Conclusions

This is the first report of SZL, a medicinal plant that grows wild in the Central Highlands of Vietnam, demonstrating inhibitory rat intestinal α -glucosidase and hypoglycemic activities. The herbal extract also demonstrated high thermal and pH stability and highly inhibits α -glucosidase from *S. cerevisiae* and *B. stearotheophilus* and α -amylases. At a dose of 100 mg/kg BW, SZL significantly reduced blood glucose in mice with no signs of diarrhea. The results in this study suggest that SZL, a potent natural source of aGIs, may be a good candidate for the treatment of type 2 diabetes. The current report also contributed to the catalog of novel anti- α -glucosidase and hypoglycemic medicinal plants.

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Authors' contributions V.B.N., S.-L.W., Q.V.N., and A.D.N. conceived the study; V.B.N. designed the study; V.B.N., Q.V.N., and T.H.N. performed the study; S.-L.W., A.D.N., V.B.N., Q.V.N., and Y.-H.K. contributed reagents/materials/analysis tools; V.B.N., S.-L.W., T.H.N., C.T.D., and T.N.T. analyzed the data; V.B.N. and S.-L.W. wrote the paper.

Compliance with ethical standards

Conflict of interest The authors declare no conflict of interest.

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